

RESEARCH DESIGN AND RESEARCH IN COMPUTING

COMM50 Postgraduate Study & Research Methods

Research Design

- Given the identified problem, we seek to obtain the 'best' possible answer(s)
 - ▣ By 'best' we usually mean the most infallible, valid, reliable, or unambiguous outcome
- Often helps to map the topic or aim and the factors that surround it
- Although simplistic, it can be thought of as a way of choosing the best approach and method(s) and planning the research accordingly
- In practice, the design will also be influenced by the research team's experience

Research in Computing



- Computing is a multi-disciplinary domain
 - ▣ Strong foundations in mathematics, logic and engineering
 - ▣ It has evolved to have relevant groundwork in psychology, usability, ubiquity, education, etc.
 - ▣ Its ubiquity provides scope for multiple approaches
- Fundamentally, it remains a science
- As such, scientific research prevails when researching and studying

Perspectives on Scientific Research

- *“Science is nothing but trained and organized common sense, differing from the latter only as a veteran may differ from a raw recruit: and its methods differ from those of common sense only as far as the guardsman's cut and thrust differ from the manner in which a savage wields his club.” -- Thomas H. Huxley*
- *“Science is nothing but developed perception, interpreted intent, common sense rounded out and minutely articulated.” -- George Santayana*

Perspectives on Scientific Research

- *“It is a good morning exercise for a research scientist to discard a pet hypothesis every day before breakfast. It keeps him young.”* -- Konrad Lorenz
- *“Basic research is what I am doing when I don't know what I am doing.”* -- Wernher von Braun
- *“Research is the process of going up alleys to see if they are blind.”* -- Marston Bates
- *“The outcome of any serious research can only be to make two questions grow where only one grew before.”* -- Thorstein Veblen

Design and Plan a Research Study



- Define the problem(s)
- What resources are available?
- Describe the intervention
- Define a sample and experimental environment
- Identify control and indicators
- Ensure fairness and transparency
- Utilise best tools for analysis and reporting

Design and Plan a Research Study



- What am I trying to do?
- How exploratory is my research?
 - ▣ Think about the amount of unknowns
 - ▣ *What? Where? Which? Why? When?*
- How applied is my research?
 - ▣ Does this have to be implemented?
- How feasible/practical is my proposed approach?
 - ▣ How much will it cost?
 - ▣ How long will it take?

Aims / Hypotheses

- Formally state the purpose of the proposed work
 - ▣ use aims and/or hypotheses
 - ▣ one paragraph
 - ▣ ... or bullet points
- Null Hypothesis (usually H_0)
 - ▣ There is no relationship between the variables being analysed (any observed changes are the result of chance)
- Alternative Hypothesis (usually H_1)
 - ▣ There is a relationship between the variables being analysed (the relationship is causal and significant)

Aims / Hypotheses

- Consider the SMART criteria when formulating aims and hypotheses:
 - Specific
 - Measureable
 - Attainable
 - Relevant
 - Time-bound



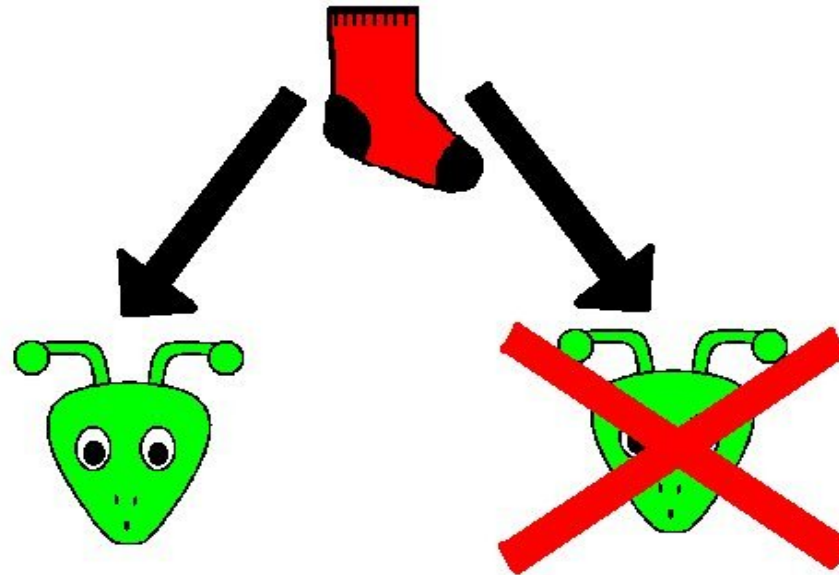
Aims / Hypotheses

- Consider the FINER (Hulley *et al.* 2007) criteria when formulating aims and hypotheses:

FINER Research Question Criteria	
F	Feasible Can you do this? Is there enough time? Do you have access to the resources and samples required?
I	Interesting Who is the research about? Who will be impacted by the research?
N	Novel What is new or different about your approach? What will you seek to demonstrate in this research?
E	Ethical Is the research ethically allowable in your institution? Who could be at risk? Does the scope need adjusting to make it compliant?
R	Relevant Will this work expand what is already known about the topic? Does it have application?

Hypotheses: A silly example!

Q. Where have all my socks gone?



Alternate Hypothesis

=

Extra-terrestrial beings have transported themselves into my house in order to steal my socks.

Null Hypothesis

=

Aliens are not to blame. There is some other explanation for the disappearing socks.

Scientific Report Structure

- Aim
 - What are you looking for in the experiment? State it.
- Experimental procedures / Methodology
 - ▣ What did you do? How was equipment set-up? What was variable and what was constant?
- Results
 - ▣ Tabulate data. Provide essential experimental conditions, and observations. Plot graphs if necessary, with explanation and legends.
- Calculation and Analysis
 - ▣ Show any calculations leading to results. Provide analysis significant to the experiment.
- Conclusion
 - ▣ State & discuss finding(s), compare with theory. Apply original thinking to state your opinion about the experiment.
- Error Analysis
 - ▣ Discuss the accuracy of your results and factors affecting them. If possible, provide suggestions to improve results for future exp't.

Experimental Variables

- Dependant Variable

- Observed and recorded during an experiment
- Not controlled by the researcher

- Independent Variable

- A variable or factor that will be manipulated in an experiment
- See if this intervention has an effect on the dependant variable

- Extraneous Variables

- Any factors that may be responsible for the behaviour observed
- Potentially contribute noise or error to the experiment, but should not be enough to bias results
- Randomisation can 'balance' the error between experimental groups
- Control ensures the influence of the EV is constant (or eliminated)

Research in Computing: Themes



- Looking for *optimal* solutions
- Monitoring *behaviour* under conditions
- *Describing* or *predicting* situations
- *Exploring* new technologies
- Performing *comparisons* between objects
- Searching for *performance* differences
 - ▣ *Efficiency*
 - ▣ *Accuracy*
 - ▣ *Quality*

Research in Computing: Methods

- Desk-based
 - ▣ Propositional
 - ▣ Literature review
 - ▣ Systematic review
- Passive
 - ▣ Case study
 - ▣ Observational
- Active
 - ▣ Evaluation
 - ▣ Experimental

*Secondary
research*

*Primary
research*

Research in Computing: Methods



- Theoretical/proof
- Case study
- Action research
- Ethnography / observation
- Simulation
- Product-based solution

Research in Computing: Methods



- Formulation of a proof
 - ▣ Analyses a problem of hypothesis from a strongly theoretical perspective
 - ▣ Can describe a solution that is valid for all (or some) cases of the problem variables
 - ▣ Simulation / modelling can be involved (but not always)
 - ▣ Valid proofs can be difficult to implement in the real-world
 - ▣ Quantitative

Research in Computing: Methods



- Empirical research
 - ▣ The traditional, structured scientific approach
 - ▣ Start with a hypothesis, devise and conduct experiments, then analyse the resultant data
 - ▣ Employs controlled experiments and statistical analysis
 - ▣ Adapting a solution to the real-world can be challenging due to uncontrollable variables / conditions
 - ▣ Usually quantitative

Research in Computing: Methods



- Product-based research
 - ▣ Concerned with building or enhancing a solution or product
 - ▣ Hands-on approach
 - ▣ Frequently encountered in IS and software development fields
 - ▣ If the problem cannot be solved, it can be difficult to describe why
 - ▣ Hard to form a widely-fitting generalisation from a single instance of a product
 - ▣ Quantitative or qualitative

Research in Computing: Methods



- 'Real-world' research
 - ▣ Analyses a solution or product in its final environment
 - ▣ Arguably, this is the only true way to test an applied solution
 - ▣ Exposes the product to a myriad of challenges and variables
 - ▣ Difficult to fully assess the success of a solution
 - ▣ Almost impossible to infer relationships and attribute causality
 - ▣ Quantitative or qualitative

Should Computer Scientists Experiment More?

- In his 1998 work, published in *IEEE Computer* magazine, Walter Tichy discusses the role of experimental research in computer science
- Tichy acknowledges that computing research stems from an engineering/mathematical research base where “proof” is all
- Tichy also considers that contemporary (c.1998) practice has moved beyond this, but now ignores scientific investigation

Should Computer Scientists Experiment More?

- Experimentation can help build a reliable base of knowledge and thus reduce uncertainty about which theories, methods, and tools are adequate.
- Observation and experimentation can lead to new, useful, and unexpected insights and open whole new areas of investigation. Experimentation can push into unknown areas where engineering progresses slowly, if at all.
- Experimentation can accelerate progress by quickly eliminating fruitless approaches, erroneous assumptions, and fads. It also helps orient engineering and theory in promising directions.

Source: Tichy, W.F., (1998). "Should Computer Scientists Experiment More?" *IEEE Computer*, vol. 31, no. 5, pp. 32-40.

Should Computer Scientists Experiment More?

- He addresses some common “excuses” given for poor practice and weak research activity in the field
- These are often cited by apprentice researchers and those lacking confidence
- Tichy introduces us to 8 fallacies of research in computer science
- Whilst some years old, his fundamental postulates are still of value...

8 Fallacies of Research in Computing (Tichy 1998)

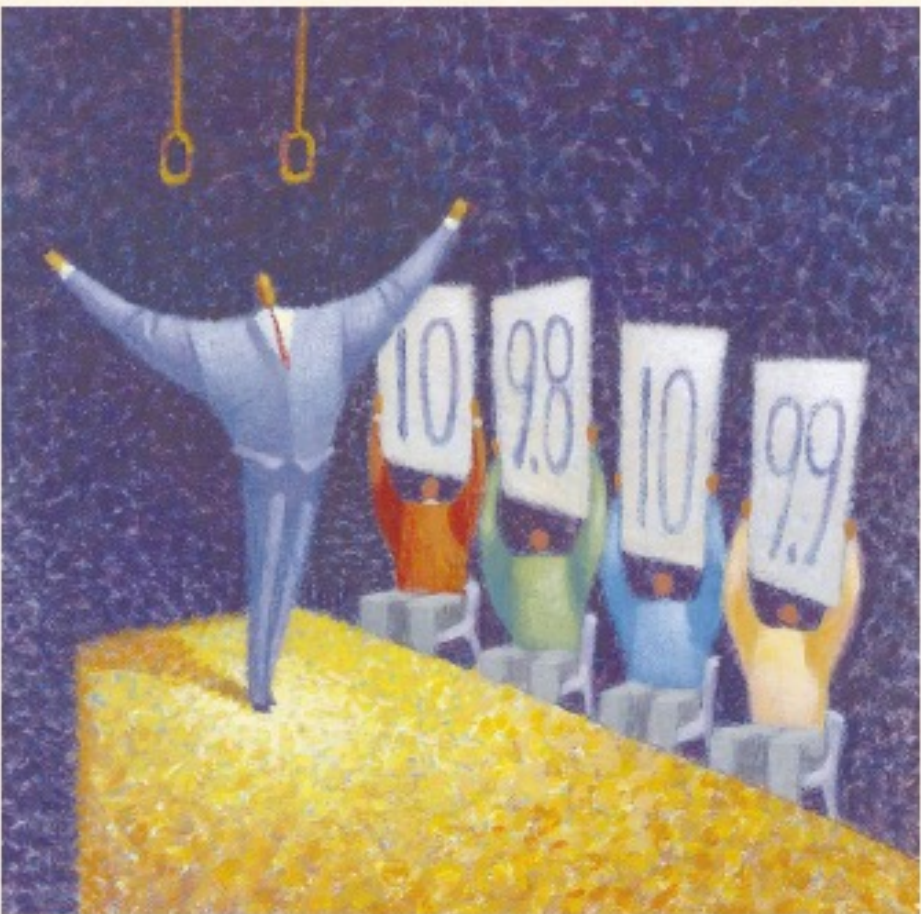
Fallacy 1. Traditional scientific method isn't applicable.



Rebuttal: To understand information processes, computer scientists must observe phenomena, formulate explanations, and test them. This *is* the scientific method.

8 Fallacies of Research in Computing (Tichy 1998)

Fallacy 2. The current level of experimentation is good enough.



Rebuttal: Relative to other sciences, the data shows that computer scientists validate a smaller percentage of their claims.

8 Fallacies of Research in Computing (Tichy 1998)

Fallacy 3. Experiments cost too much.



Rebuttal: Meaningful experiments can fit into small budgets; expensive experiments can be worth more than their cost.

Fallacy 4. Demonstrations will suffice.

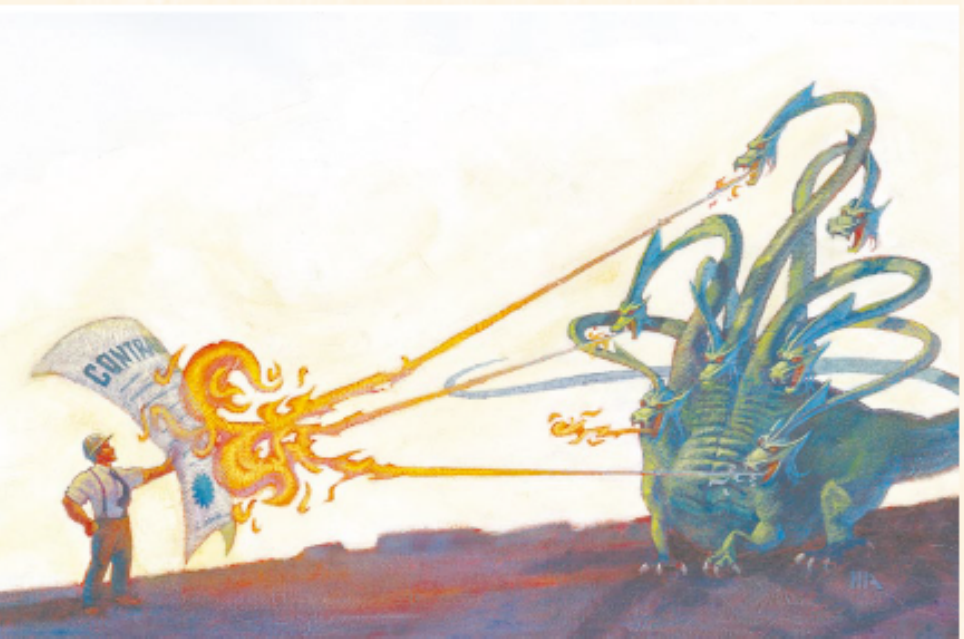


Rebuttal: Demos can provide incentives to study a question further. Too often, however, these demos merely illustrate a potential.

8 Fallacies of Research in
Computing (Tichy 1998)

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Fallacy 5. There's too much noise in the way.



Rebuttal: Fortunately, benchmarking can be used to simplify variables and answer questions.

8 Fallacies of Research in Computing (Tichy 1998)

Fallacy 6. Experimentation will slow progress.



Rebuttal: Increasing the ratio of papers with meaningful validation has a good chance of actually accelerating progress.

Fallacy 7. Technology changes too fast.



Rebuttal: If a question becomes irrelevant quickly, it is too narrowly defined and not worth spending a lot of effort on.

8 Fallacies of Research in Computing (Tichy 1998)

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Fallacy 8. You'll never get it published.

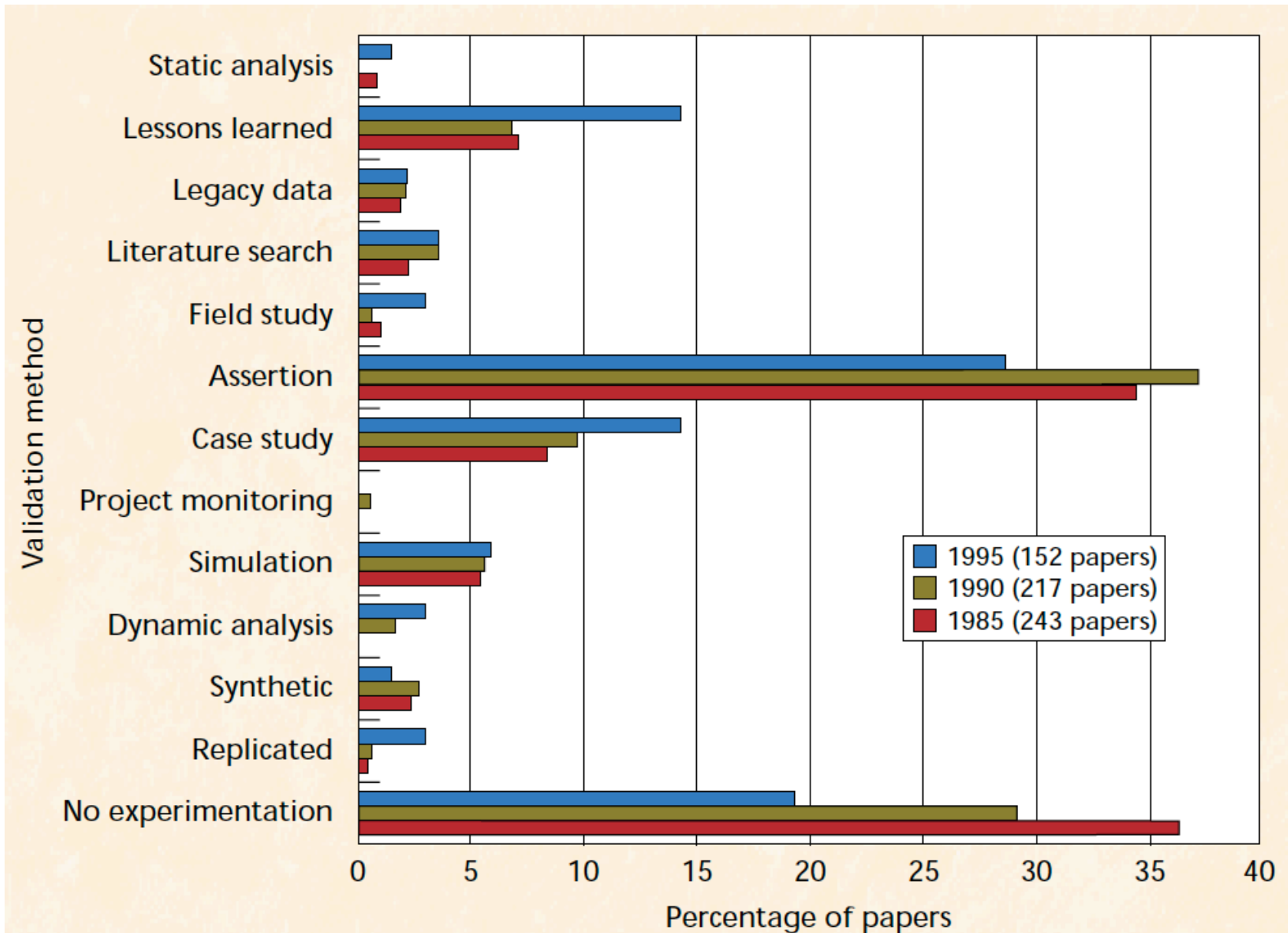


Rebuttal: Smaller steps are still worth publishing because they improve our understanding and raise new questions.

Scientific Method in Computing

- *“Scientists should create models, formulate hypotheses, and test them using experiments.”*
(Tichy, 1998)
- In a work produced by Sjøberg et al. (2005), on software engineering, only 1.5% of published works surveyed employed a controlled experiment
 - ▣ This paper is a great read in itself; exemplifying a large-scale study with a defined methodology and aim

“...computer science has not developed a concise taxonomy of methods for demonstrating the validity of new techniques” (Zelkowitz & Wallace, 1998)



Zelkowitz, M.V.; Wallace, D.R., "Experimental models for validating technology," *Computer*, vol. 31, no.5, pp.23,31, May 1998

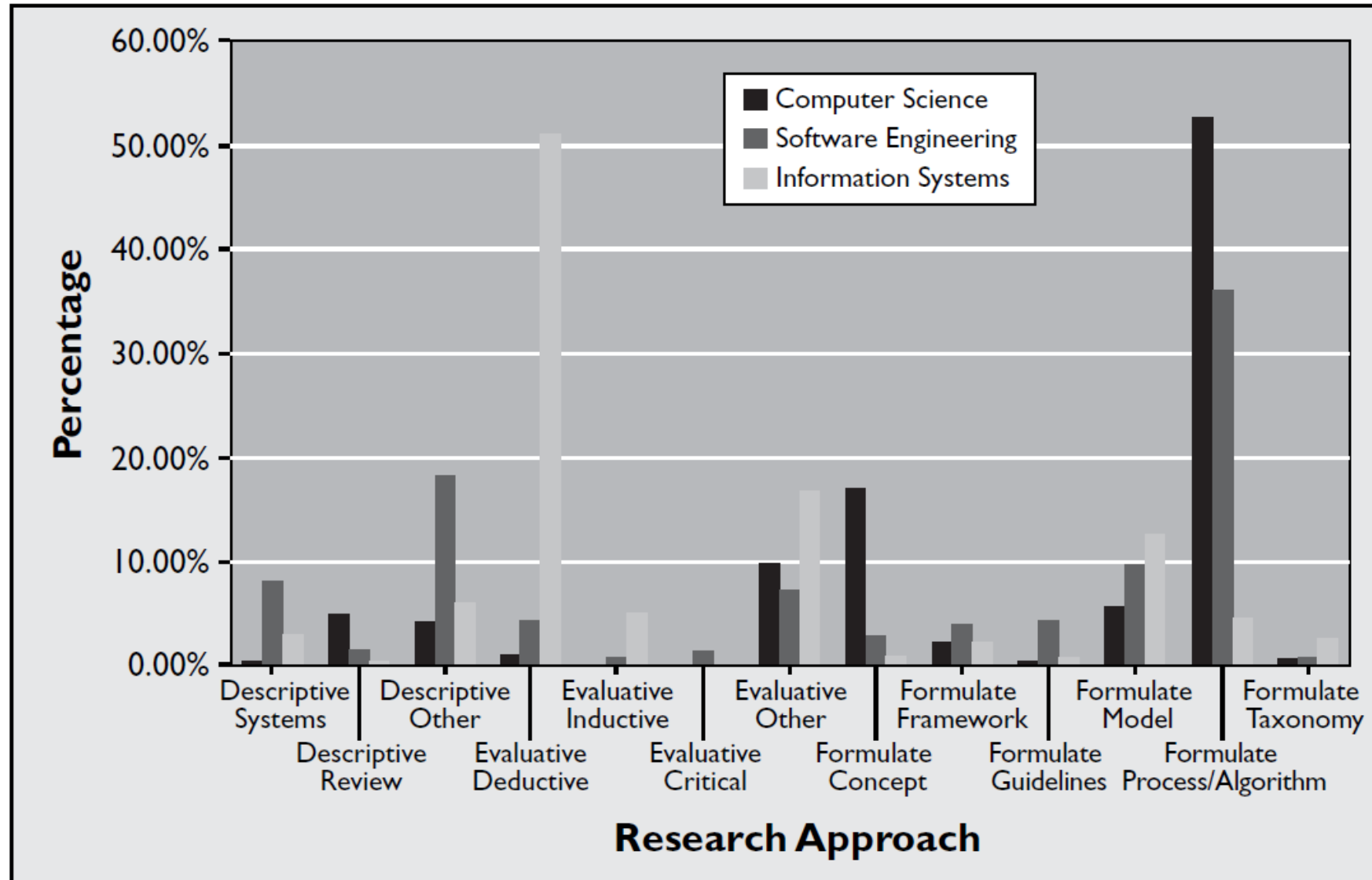
“...computer science has not developed a concise taxonomy of methods for demonstrating the validity of new techniques” (Zelkowitz & Wallace, 1998)

Data Collection Example

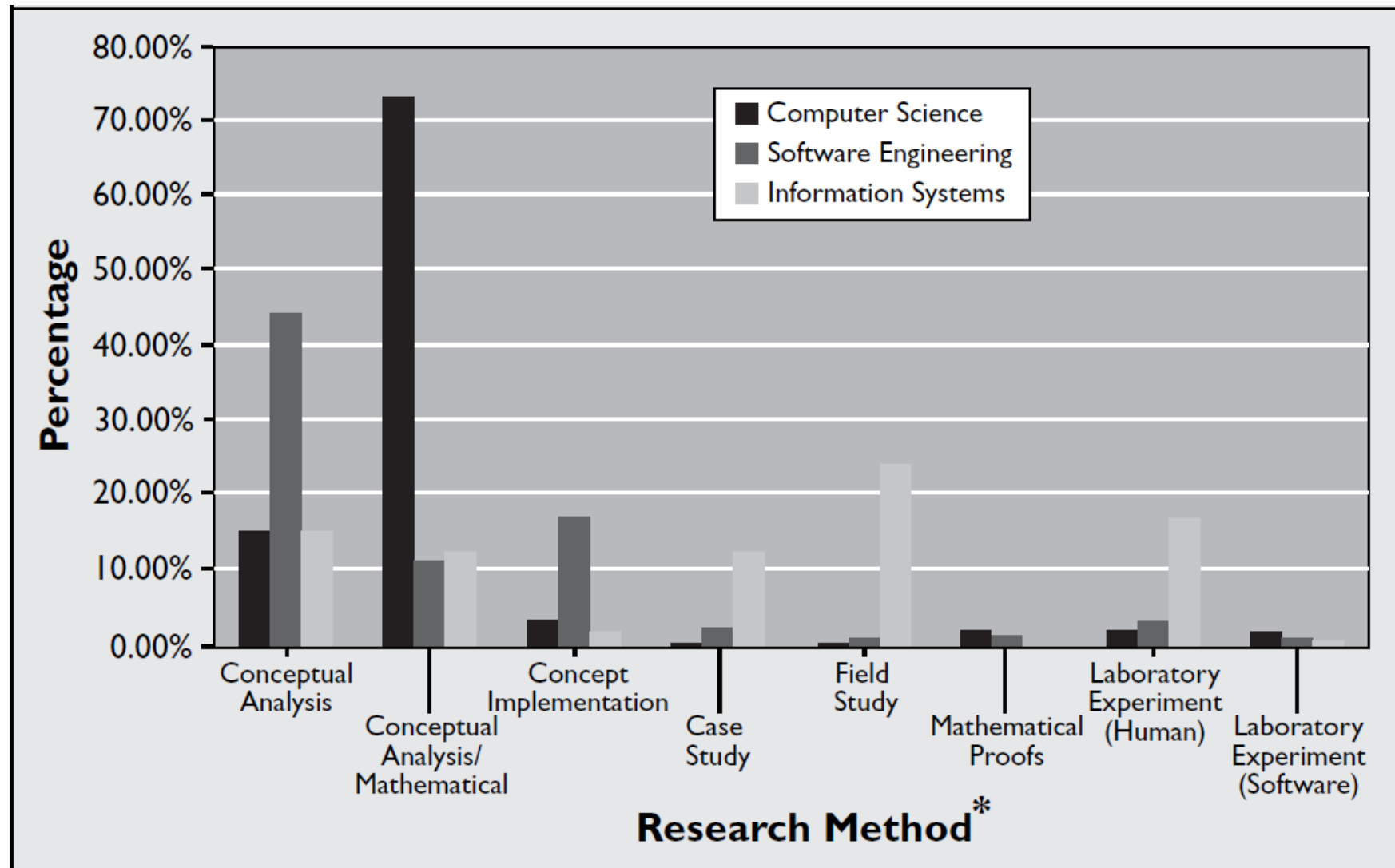
Several evaluation methods can be used to test a new tool:

- *Project monitoring.* Use the new tool in a project and collect the usual accounting data from the project.
- *Case study.* Use the tool as part of new development. Collect data to determine whether the developed product is easier to produce than similar projects in the past.
- *Assertion.* Use the tool to test a simple 100-line program to show that it can find all errors.
- *Field study.* Distribute the tool across several projects. Collect data on the tool's impact.
- *Literature search.* Find other published studies that analyze the behavior of similar tools.
- *Legacy data.* Find a previously completed project that collected data on using the tool. Analyze this data to see if tool was effective.
- *Lessons learned.* Find a completed project that used this tool and interview participants to see if tool had an impact on the project.
- *Static analysis.* Use a control flow analysis tool to see if one design method results in fewer logic errors than another design method.
- *Replicated experiment.* Develop multiple instances of a module using and not using the tool. Measure the differences.
- *Synthetic.* Have 20 programmers spend two hours trying to debug a module, half of them using the tool and half of them using other techniques.
- *Dynamic analysis.* Execute a program with a new algorithm and compare its performance with the earlier version of the program.
- *Simulation.* Generate a set of data points randomly and then execute the tool alongside another tool to determine effectiveness in finding errors in a given module.

Computing Approaches and Methods (Glass *et al.* 2004)

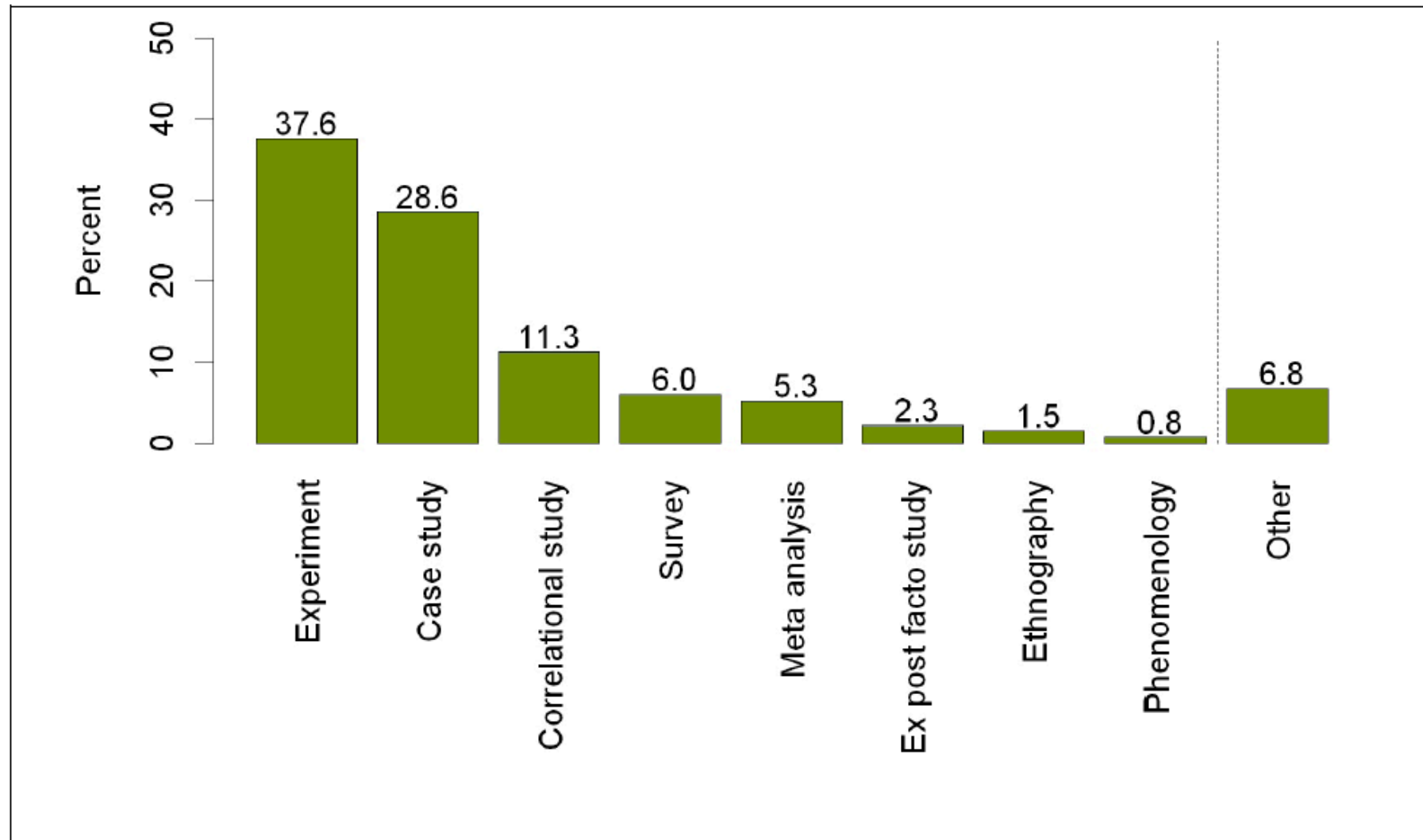


Computing Approaches and Methods (Glass *et al.* 2004)



* Note that Figure 2 (bottom) shows only those research methods with the highest representation.

More Recently (Höfer & Tichy 2007)



More Recently (Höfer & Tichy 2007)

Method	Definition
Case study	In-depth analysis of a particular project, event, organization, etc.
Correlational study	Measuring variables and determining the degree of relationship that exists between them.
Ethnography	Description and interpretation of the culture of a group of people.
Ex post facto study	Study in which the variables of interest are not subject to direct manipulation, but must be chosen after the fact (e.g., when analyzing software repositories).
Experiment	Quantitative study to test cause-and-effect relationships.
Meta analysis	Integrates and/or describes the results of several studies.
Phenomenology	Description of an individual's or a group's experience of a phenomenon.
Survey	Data is collected by interviewing a representative sample of some population.

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